Beam-Steered Single-LED Single-PD Visible-Light Positioning for 3D Localization

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***Abstract*—(draft) This paper investigates a practical three-dimensional visible-light positioning (VLP) architecture in which a single ceiling-mounted LED transmitter sequentially steers its optical beam towards K distinct orientations, while a single upward-looking photodiode (PD) placed at an unknown position gathers N power samples per orientation. Two successive estimation stages are proposed: (i) direction finding of the LED–PD line-of-sight, followed by (ii) range recovery through a dedicated beam-steering orientation that maximises the received signal-to-noise ratio (SNR). We derive the Cramér–Rao lower bound, here expressed as the position-error bound (PEB), to benchmark any unbiased estimator. A genetic-algorithm optimiser is then designed to compute the K-orientation sets that minimise the 90th percentile of the PEB over a cuboidal testbed. Three estimators are analysed: a statistically efficient but iterative non-linear (NL) estimator, a closed-form generalised least-squares (GLS) estimator, and a computationally lightweight weighted least-squares (WLS) approximation. Extensive simulations corroborate the theoretical bound and reveal that the GLS reaches near-optimal performance with two orders of magnitude lower latency than NL, whereas WLS attains within 1 dB of GLS for typical indoor geometries. Guidelines are finally provided to select K and the Lambertian half-power angle $\cancel{\Phi_{1/2}}$, trading off latency against accuracy.**

***Index Terms*—Visible-light positioning, single-LED VLP, Cramér–Rao lower bound, 3-D localisation, generalised least squares.**

I. Introduccion

[1]–[16]

II. System Model

Figure 1 sketches the single-LED visible-light positioning (VLP) setup analysed in this work. A light-emitting diode (LED) transmitter is mounted on the ceiling at a known position $\mathbf{T} = (0, 0, H)^\mathsf{T}$ and can steer its optical axis along a unit vector $\mathbf{n}_t^{(i)} \in \mathbb{S}^2$ for the i-th orientation, $i = 1, \ldots, K$. The receiver is a single photodiode with fixed vertical axis $\mathbf{n}_r = (0, 0, 1)^\mathsf{T}$ located at an unknown position $\mathbf{R} = (x, y, z)^\mathsf{T}$. Let $\mathbf{d} = \mathbf{R} - \mathbf{T}$ be the transmitter-to-receiver displacement and $\mathbf{n}_d = \mathbf{d}/\|\mathbf{d}\|$ its direction.

For the i-th LED orientation, the optical power collected by the detector is

$$P_r^{(i)} = P_t\, h_{\text{LOS}}^{(i)} + \eta^{(i)}, \tag{1}$$

where P_t is the transmitted radiant power, $\eta^{(i)} \sim \mathcal{N}(0, \sigma^2)$ is zero-mean AWGN, and $h_{\text{LOS}}^{(i)}$ is the line-of-sight (LOS) channel gain modelled by a Lambertian source:

$$h_{\text{LOS}}^{(i)} = \begin{cases} \dfrac{(m+1)A_{\text{det}}}{2\pi\|\mathbf{d}\|^2} \big(\cos\phi_i\big)^m \cos\psi, & \psi \le \Psi_{\text{FOV}}, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

with

$$m = -\frac{\ln 2}{\ln\!\big(\cos\Phi_{1/2}\big)}$$

~~the Lambertian order corresponding to half-power semi-angle $\Phi_{1/2}$~~, A_{det} the photodiode effective area, Ψ_{FOV} the receiver field of view, and the irradiance/incidence cosines

$$\cos\phi_i = \frac{\mathbf{n}_t^{(i)} \cdot \mathbf{d}}{\|\mathbf{d}\|}, \qquad \cos\psi = \frac{\mathbf{n}_r \cdot (-\mathbf{d})}{\|\mathbf{d}\|}. \tag{3}$$

At every orientation i the receiver averages N samples, producing the unbiased estimator

$$\bar{P}_r^{(i)} = \frac{1}{N} \sum_{j=1}^{N} P_{r,j}^{(i)} \sim \mathcal{N}\big(P_r^{(i)}, \sigma^2/N\big). \tag{4}$$

The corresponding instantaneous signal-to-noise ratio (SNR) and its test-bed average are

$$\text{SNR}_r^{(i)} = \frac{P_t\, h_{\text{LOS}}^{(i)}}{\sigma^2}, \qquad \overline{\text{SNR}}_r^{(i)} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \text{SNR}_r^{(i)},$$

where $\mathcal{R}$ denotes the set of receiver test-bed positions. Results are reported in decibels through $10 \log_{10}\big(\overline{\text{SNR}}_r^{(i)}\big)$.

A. General 3-D Single-LED Positioning Procedure

The proposed localisation pipeline comprises two consecutive stages.

1) Direction finding: The first stage estimates the direction vector $\mathbf{n}_d$ by steering the LED through the K predefined orientations $\{\mathbf{n}_t^{(i)}\}_{i=1}^K$ and processing the power means $\{\bar{P}_r^{(i)}\}$. Section V (non-linear estimator) and Section VI (linear GLS/WLS estimators) detail statistically grounded solutions.

2) Distance recovery: After direction finding, the LED is aligned with the estimated direction $\hat{\mathbf{n}}_d$ and, for beam-forming symmetry, the receiver is re-oriented to $\mathbf{n}_r = -\hat{\mathbf{n}}_d$. Under this beam-steered configuration the received power obeys

$$\mu^* = \frac{C}{\|\mathbf{d}\|^2}, \qquad C = \frac{P_t(m+1)A_{\text{det}}}{2\pi}, \tag{5}$$

where C is an optical constant calibrated once per hardware set-up. Collecting N independent measurements $\{\mu_j^*\}_{j=1}^N$ and

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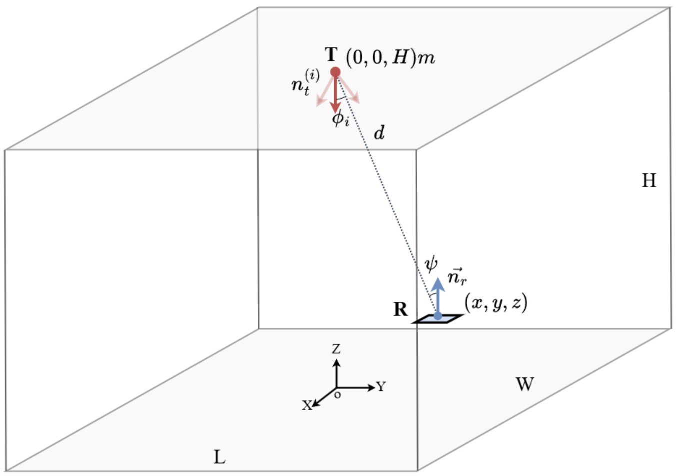


Fig. 1. Configuration of the system Single LED Visible Light Positioning.

using (4) yields the MVU estimate $\hat{\mu}^* \sim \mathcal{N}(\mu^*, \sigma^2/N)$. Substituting $\hat{\mu}^*$ into (5) provides the range estimate

$$\hat{d} = \sqrt{\frac{C}{\hat{\mu}^*}}, \tag{6}$$

and, finally, the 3-D position estimate

$$\hat{\mathbf{R}} = \mathbf{T} + \hat{d}\,\hat{\mathbf{n}}_d. \tag{7}$$

III. Position Error Bound

This section derives the position error bound (PEB), i.e., the Cramér–Rao lower bound (CRLB) specialised to the receiver's three-dimensional coordinate vector $\mathbf{R} = (x, y, z)^\mathsf{T}$. The PEB quantifies the minimum achievable variance of any unbiased estimator of $\mathbf{R}$ under the proposed VLP model and serves both as a benchmark for estimator performance and as an objective for system design.

A. Joint statistical model of the observations

According to Sec.II, the sample-mean optical power for each of the K transmitter orientations satisfies

$$\bar{P}_r^{(i)} \sim \mathcal{N}\big(\mu_i(\mathbf{R}),\, \sigma^2/N\big), \quad i = 1, \ldots, K,$$

where $\mu_i(\mathbf{R}) = P_t\, h_{\mathrm{LOS}}^{(i)}(\mathbf{R})$ and $h_{\mathrm{LOS}}^{(i)}$ is defined in (2) with the cosines in (3). Following beam-steering, the range-recovery statistic obeys

$$\mu^* \sim \mathcal{N}\big(C/d^2(\mathbf{R}),\, \sigma^2/N\big),$$

where $d(\mathbf{R}) = \|\mathbf{R} - \mathbf{T}\|$ and C is the optical constant introduced in (5).

Stacking these $K+1$ independent observations into $\mathbf{z} = [\,\bar{P}_r^{(1)}, \ldots, \bar{P}_r^{(K)}, \mu^*\,]^\mathsf{T}$ yields the multivariate Gaussian model

$$\mathbf{z} \sim \mathcal{N}\big(\boldsymbol{\mu}(\mathbf{R}),\, (\sigma^2/N)\,\mathbf{I}_{K+1}\big),$$

with mean vector $\boldsymbol{\mu}(\mathbf{R}) = [\mu_1(\mathbf{R}), \ldots, \mu_{K+1}(\mathbf{R})\,]^\mathsf{T}$ and diagonal covariance σ^2/N.

B. Log-likelihood and score function

Under the assumption of independent Gaussian measurements, the log-likelihood of $\mathbf{R}$ given $\mathbf{z}$ is

$$\ell(\mathbf{R}) = \ln L(\mathbf{R} \mid \mathbf{z}) = -\frac{N}{2\sigma^2} \sum_{i=1}^{K+1} \big[z_i - \mu_i(\mathbf{R})\big]^2 + \text{constant}. \tag{8}$$

Differentiating $\ell(\mathbf{R})$ with respect to $\mathbf{R}$ defines the score vector

$$\mathbf{s}(\mathbf{R}) = \nabla_{\mathbf{R}}\, \ell(\mathbf{R}) = \frac{N}{\sigma^2} \sum_{i=1}^{K+1} \big[z_i - \mu_i(\mathbf{R})\big]\, \nabla_{\mathbf{R}}\, \mu_i(\mathbf{R}),$$

where $\nabla_{\mathbf{R}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^\mathsf{T}$.

C. Fisher Information Matrix

For distributions in the exponential family, such as the homoscedastic Gaussian model adopted here, the Fisher Information Matrix (FIM) can be obtained as the variance of the score vector, avoiding second-order differentiation and yielding greater numerical stability:

$$\mathbf{I}(\mathbf{R}) = \mathrm{Var}\big[\mathbf{s}(\mathbf{R})\big] = \mathbb{E}_{\mathbf{z}}\big[\mathbf{s}(\mathbf{R})\,\mathbf{s}(\mathbf{R})^\mathsf{T}\big] \tag{9}$$

$$= \frac{N}{\sigma^2} \sum_{i=1}^{K+1} \big[\nabla_{\mathbf{R}}\mu_i(\mathbf{R})\big]\, \big[\nabla_{\mathbf{R}}\mu_i(\mathbf{R})\big]^\mathsf{T}. \tag{10}$$

The second equality follows from $\mathbb{E}[z_i] = \mu_i(\mathbf{R})$, which implies $\mathbb{E}\big[(z_i - \mu_i)(z_j - \mu_j)\big] = 0$ for $i \neq j$. Hence the total information is the coherent superposition of the differential contributions from each LED orientation and the range measurement. Required gradients are:

1) Orientation measurements:
For the i-th orientation

$$\nabla_{\mathbf{R}}\mu_i(\mathbf{R}) = P_t\, \nabla_{\mathbf{R}} h_{\mathrm{LOS}}^{(i)}(\mathbf{R}),$$

which, by differentiating (2) via the chain rule on $d(\mathbf{R})$, $\cos\phi_i$ and $\cos\psi$, yields the closed-form

$$\nabla_{\mathbf{R}}\mu_i(\mathbf{R}) = \frac{C}{d^3} \Big[m \cos^{m-1}\phi_i\, \cos\psi\, \mathbf{n}_t^{(i)} - \cos^m\phi_i\, \mathbf{n}_r - (m+3)\cos^m\phi_i\, \cos\psi\, \mathbf{n}_d \Big], \tag{11}$$

which vanishes whenever $\psi > \Psi_{\mathrm{FOV}}$ since $h_{\mathrm{LOS}}^{(i)} = 0$ outside the receiver's field of view.

2) Distance-recovery measurement:

$$\nabla_{\mathbf{R}}\mu_{K+1}(\mathbf{R}) = -\frac{2C}{d^4}\big(\mathbf{R} - \mathbf{T}\big) = -\frac{2C}{d^3}\, \mathbf{n}_d, \tag{12}$$

where $\mathbf{n}_d$ is the unit displacement vector defined in Section II.

D. Formal definition of the PEB

Let $\hat{\mathbf{R}}$ be any unbiased estimator of $\mathbf{R}$. The Cramér–Rao inequality asserts

$$\mathrm{Cov}\big[\hat{\mathbf{R}}\big] \succeq \mathbf{I}^{-1}(\mathbf{R}).$$

Accordingly, a scalar performance metric, the position error bound, is defined as

$$\mathrm{PEB}(\mathbf{R}) = \sqrt{\mathrm{tr}\big\{\mathbf{I}^{-1}(\mathbf{R})\big\}}. \tag{13}$$

The square root of the trace of $\mathbf{I}^{-1}$ corresponds to the minimum achievable root-mean-square error (RMSE) in position estimation for the described VLP configuration. A small PEB indicates that the chosen set of LED orientations and the range measurement provide sufficient and complementary information for high-precision localisation; conversely, a large PEB suggests the need to increase K, improve the signal-to-noise ratio, or redesign the orientation pattern.

E. Number of LED Orientations

a) Underdetermined regime ($K \leq 2$): In this regime, the Fisher information matrix $\mathbf{I}(\mathbf{R})$ from (10) is rank-deficient. Each of the K orientation measurements contributes at most one independent gradient vector, and the distance-recovery measurement adds a single additional vector. When $K \leq 2$, there are fewer than three linearly independent gradients in total, so $\mathbf{I}(\mathbf{R})$ is singular and the CRLB on any unbiased position estimate diverges [15].

b) Just-determined regime ($K = 3$): With three distinct, non-coplanar LED orientations plus the range measurement, the total of four gradient vectors can span the three-dimensional parameter space. In generic geometries the FIM is full rank and $\mathbf{I}^{-1}(\mathbf{R})$ exists, yielding a finite PEB. However, the absence of redundancy makes the bound highly sensitive to small perturbations in orientation geometry [2].

c) Overdetermined regime ($K \geq 4$): For four or more orientations, the FIM accumulates redundant outer- products of gradients. This overdetermination improves the numerical conditioning of $\mathbf{I}(\mathbf{R})$ and generally lowers the PEB, since extra measurements provide additional independent information beyond the minimal requirement [4], [16].

IV. Optimization of the Orientation Set

Equation (11) shows that the Fisher information matrix $\mathbf{I}(\mathbf{R})$ depends explicitly on the transmitter orientation vectors $\{\mathbf{n}_t^{(i)}\}_{i=1}^{K}$, and hence so does the PEB in (13). Accordingly, selecting the orientation set that minimises the PEB is essential for optimal localisation accuracy. Since $\mathrm{PEB}(\{\mathbf{n}_t^{(i)}\})$ is non-convex in the orientation parameters, direct analytical minimisation is intractable.

a) Genetic algorithm framework: To address this, we employ a genetic algorithm (GA) to search for the orientation ensemble that minimises the global PEB for a given K. Each GA individual encodes K steering directions via elevation–azimuth pairs $\{(\theta_i, \rho_i)\}_{i=1}^{K}$, yielding a $2K$-dimensional genotype $\mathbf{x} = [\theta_1, \rho_1, \ldots, \theta_K, \rho_K]^{\mathsf{T}}$. The mapping to unit-norm orientation vectors is

$$\mathbf{n}_t^{(i)} = \big(\sin\theta_i \cos\rho_i,\ \sin\theta_i \sin\rho_i,\ -\cos\theta_i\big).$$

The GA's fitness metric is defined as the RMSE of the PEB over all 3D grid points spanning the testbed, computed by evaluating $\mathrm{PEB}(\mathbf{R})$ from (13). Standard genetic operators such as tournament selection, crossover probability p_c, and mutation rate p_m, are applied over G generations. The algorithm is implemented in MATLAB; optimization parameters are summarised in Table I.

TABLE I: Parameters used for genetic algorithm optimization

System Parameters	
Parameter	**Value**
Transmitter position	$T = (0, 0, 2)$ m
Transmitted optical power	$P_t = 0.405$ W
Half-power semi-angle	$\Phi_{1/2} = 45°$
Lambertian order	$m = \cancel{-\ln 2/\ln(\cos 45°) =} 2$
Photodiode effective area	$A_{\text{det}} = 4.8 \times 5.5\,\mathrm{mm}^2$
Receiver field of view (FOV)	$\Psi_{\text{FOV}} = 85°$
Noise variance per sample	$\sigma^2 = 10^{-21} \times 30 \times 10^6\,\mathrm{W}^2$
Samples per orientation	$N = 1000$
Genetic Algorithm	
Parameter	**Value**
Initial population	300
Maximum generations	200
Crossover rate	0.8
Search range	$\theta_i \in [0, 80°],\ \rho_i \in [0, 360°]$
Optimization metric	RMSE
Decision variables	$[\theta_1, \rho_1, \ldots, \theta_K, \rho_K]$
3D Test-bed	
Parameter	**Value**
Room dimensions (L×W×H)	$3 \times 3 \times 2\,\mathrm{m}^3$
Range in x, y	$[-1.5,\ 1.5]$ m
Range in z	$[0,\ 1.2]$ m
Grid step	0.2 m

Table II lists the optimal K-orientation sets for the system parameters in Table I. Each set is unique and no set is a subset of another. For every optimal orientation ensemble, we compute the PEB (or equivalently the RMS error) at each receiver location within the testbed grid. In Figure 2, the box-and-whisker plots for the optimal sets are shown in green, alongside those for randomly chosen orientations. The optimal sets yield consistently lower statistical measures (median, interquartile range) than their random counterparts, and the median RMS decreases as K increases.

Figure 3 presents spatial heat maps of the PEB across a $3 \times$ 3m floor at $z =$ 0.8m for $K = 5$ orientations with the transmitter at the room centre. In the optimal configuration (Fig. 3a), the highest errors occur uniformly along the boundaries and especially at the corners, analogous to multi-transmitter layouts, but remain spatially consistent. By contrast, the random configuration (Fig. 3b) exhibits both larger and more irregular error patterns.

Figure 4 depicts the 90th-percentile of the PEB CDF ($\mathrm{PEB}_{90\%}$) for the optimal K-orientation set under three half-power semi-angle scenarios $\Phi_{1/2}$ at an effective SNR of 14 dB. In all scenarios, $\mathrm{PEB}_{90\%}$ decreases as K increases, indicating improved localisation precision with additional orientations. A balance among latency, computational complexity, and positioning accuracy is attained at $K = 5$. For $\Phi_{1/2} = 30°$, at least

TABLE II: Optimal orientations (θ_i, ρ_i) in degrees for various K in a $3 \times 3 \times 2\,\mathrm{m}^3$ room

K	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$	$i=7$	$i=8$	$i=9$	$i=10$
3	(35.4,140.1)	(33.3,36.4)	(29.6,262.7)							
4	(38.9,90.6)	(41.5,0.1)	(41.8,180.1)	(38.8,270.2)						
5	(0.1,211.1)	(50.7,180.0)	(50.4,359.9)	(50.6,90.0)	(50.6,270.0)					
6	(17.2,306.9)	(54.5,266.1)	(22.5,140.4)	(52.2,360.0)	(52.4,84.0)	(55.8,185.2)				
7	(58.9,355.7)	(53.8,170.7)	(27.8,43.8)	(5.4,305.9)	(54.4,96.5)	(35.1,220.0)	(54.8,278.6)			
8	(51.8,89.4)	(61.5,268.3)	(27.3,317.0)	(6.5,318.3)	(57.8,5.8)	(53.7,171.3)	(38.0,200.3)	(39.3,91.1)		
9	(56.9,178.7)	(36.5,266.8)	(33.9,182.3)	(2.5,28.2)	(42.2,78.4)	(53.1,97.5)	(57.9,359.7)	(37.1,355.1)	(58.2,272.1)	
10	(56.0,3.6)	(53.2,182.5)	(54.9,356.8)	(11.9,38.1)	(61.3,270.3)	(50.2,91.3)	(47.6,174.7)	(43.4,89.4)	(32.5,277.6)	(15.1,255.3)

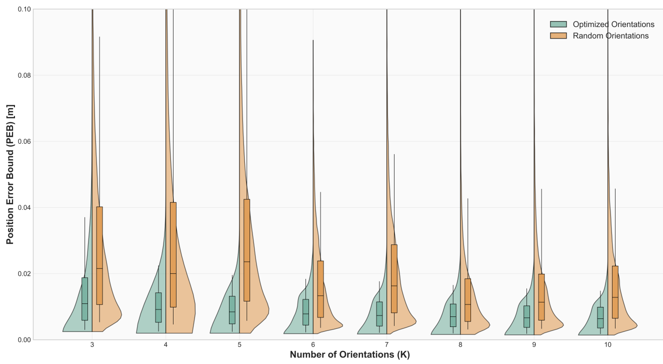
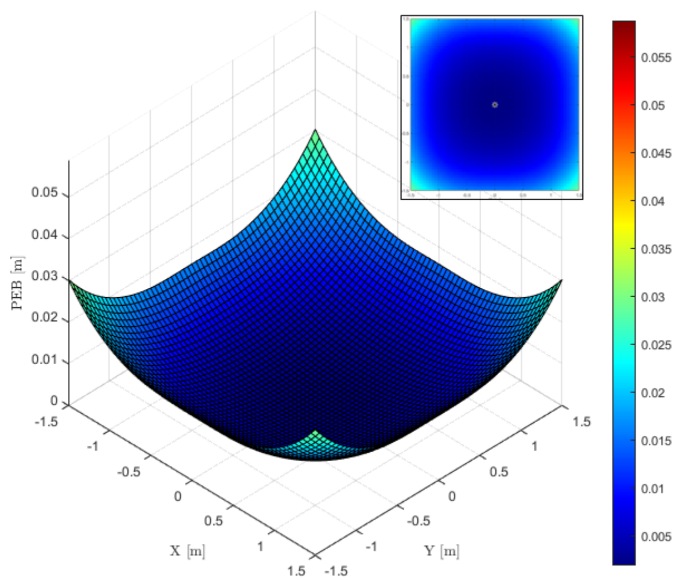
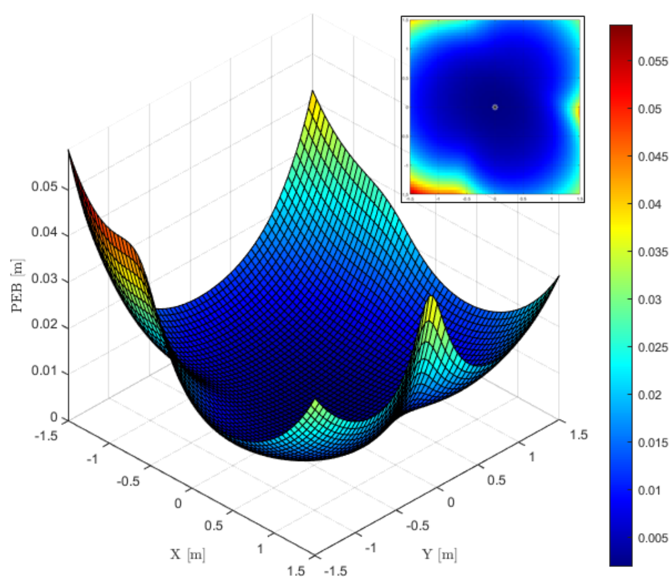


Fig. 2. RMS performance in a 3×3m testbed using an optimal set of K orientations versus a sub-optimal set of K orientations.



(a)



(b)

Fig. 3. Heat map of the PEB across a 3×3m testbed at $z = 0.8$m for $K = 5$ orientations: (a) optimal set, (b) sub-optimal set.

four orientations are necessary due to the increased directivity limiting coverage with only three orientations; configurations with $K \geq 4$ outperform alternatives. In contrast, $\Phi_{1/2} = 45°$ achieves competitive error performance starting at $K = 3$, and given its widespread use in commercial LEDs, it is adopted for the subsequent simulations.

Figure 5 illustrates the dependence of the 90th-percentile PEB ($\mathrm{PEB}_{90\%}$) on the ~~signal-to-noise ratio (SNR)~~ for various optimized K-orientation sets. As the SNR increases, $\mathrm{PEB}_{90\%}$ decreases monotonically. Moreover, at any fixed SNR, increasing K yields a lower PEB, and the spacing between the curves diminishes at high SNR, indicating reduced incremental benefit from additional orientations under noise-limited conditions.

V. Nonlinear Estimator

The optical power collected by the photodiode depends nonlinearly on its relative position to the LED, through the range d, the irradiance angle ϕ_i, and the incidence angle ψ. Denoting the receiver's orientation vector as $\mathbf{n}_r = [\alpha, \beta, \gamma]^\mathsf{T}$, we have

$$\cos\psi = \mathbf{n}_r \cdot (-\mathbf{n}_d) = -\frac{\alpha x + \beta y + \gamma\,(z - H)}{d}, \qquad (14)$$

where $\mathbf{R} = (x, y, z)^\mathsf{T}$ is the receiver position and H the LED height. Substituting (3), (14), and $d = \sqrt{x^2 + y^2 + (z-H)^2}$ into the received-power model (1), and then forming the sample mean as in (4), yields

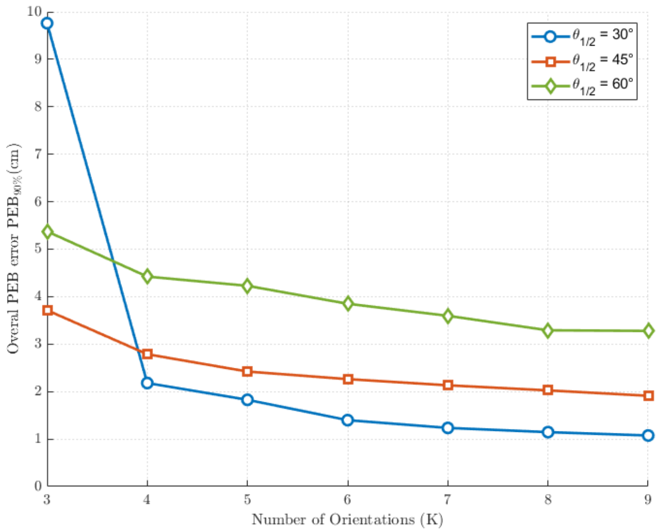


Fig. 4. Algorithm performance versus number of orientations under three half-power semi-angle ($\Phi_{1/2}$) scenarios.

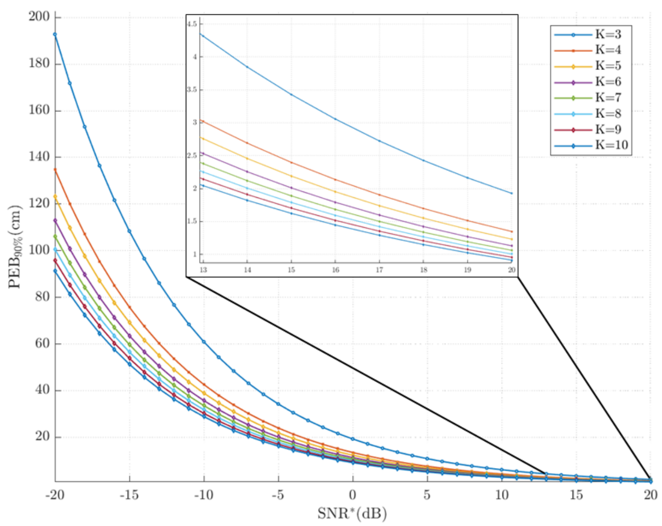


Fig. 5. Variation of the 90th-percentile PEB ($\mathrm{PEB}_{90\%}$) with SNR for different optimized K-orientation configurations.

$$P_{o,r}^{(i)} = -\,C\,\frac{\big(a_i x + b_i y + c_i(z-H)\big)^m}{\big[x^2+y^2+(z-H)^2\big]^{\frac{m+3}{2}}}\,(\alpha x+\beta y+\gamma\,(z-H))+n \tag{15}$$

where $n \sim \mathcal{N}(0, \sigma^2/N)$ and C is defined in (5). Thus, the observations relate nonlinearly to the receiver coordinates $\mathbf{R}$. To recover the direction unit vector $\mathbf{n}_d$ rather than the full coordinate triple, one imposes the constraint $x^2 + y^2 + (z-H)^2 = 1$. Defining

$$Q_i(x,y,z) = a_i x + b_i y + c_i\,(z-H), \tag{16}$$
$$L(x,y,z) = \alpha x + \beta y + \gamma\,(z-H), \tag{17}$$

the model in (15) can be compactly written as

$$P_{o,r}^{(i)} = -\,C\,Q_i^m(x,y,z)\,L(x,y,z) + n, \quad n \sim \mathcal{N}(0,\sigma^2/N).$$

This scenario exemplifies a classical estimation problem in which the averaged observations $P_{o,r}^{(i)}$ for each orientation i are nonlinearly related to the unknown receiver coordinates and corrupted by additive Gaussian noise of known variance. Although no unbiased efficient estimator exists for this non-linear model, one can derive a ~~minimum variance unbiased (MVU)~~ estimate by solving a constrained nonlinear least-squares problem.

Define the cost function

$$F(x,y,z) = \sum_{i=1}^{K} F_i(x,y,z), \tag{18}$$

where

$$F_i(x,y,z) = \big[P_{o,r}^{(i)} + C\,Q_i(x,y,z)\,L(x,y,z)\big]^2. \tag{19}$$

Here Q_i and L are as defined in (16)–(17), and C is the optical constant introduced in (5).

Since our goal is to recover the unit-norm direction vector $\mathbf{n}_d$ rather than the absolute coordinates, the variables $\big(x, y, z-H\big)$ must lie on the unit sphere $\mathbb{S}^2$. Moreover, the polynomial terms $Q_i(x,y,z)$ and $L(x,y,z)$ must satisfy

$$Q_i(x,y,z) \geq 0, \quad L(x,y,z) \leq 0,$$

corresponding respectively to $\cos\phi_i \geq 0$ and $-\cos\psi \leq 0$ for incidence and irradiance angles in $[0, \pi/2]$.

Thus, the nonlinear direction-finding estimator is

$$\hat{\mathbf{n}}_{d,\mathrm{NL}} = \arg \min_{\substack{(x,y,z-H)\in\mathbb{S}^2 \\ Q_i(x,y,z)\geq 0 \\ L(x,y,z)\leq 0}} F(x,y,z), \tag{20}$$

with $F(x,y,z)$ given by (18)–(19). In practice, this optimisation can be performed in MATLAB using the `solve` function, which internally invokes `fmincon` and employs gradient-descent algorithms suitable for constrained nonlinear problems.

VI. Lineal Estimator

The present section develops the first stage with full mathematical rigour and presents two statistically grounded estimators: (i) generalised least squares (GLS), which is minimum-variance under Gaussian noise with known covariance, and (ii) weighted least squares (WLS), an inexpensive approximation that neglects the weak cross–correlation between equations.

A. Direction Estimation via GLS

This subsection consolidates the analytical pipeline that converts the raw optical powers into a closed-form, statistically efficient estimate of the direction vector $\mathbf{n}_d$. The exposition proceeds from the deterministic constraints, through a rigorous stochastic characterisation of the residuals, to the final GLS solution.

1) Lambertian power ratios and homogeneous constraints: Let $\mu_i = \mathbb{E}[P_r^{(i)}]$ be the noise-free power at orientation i. We define power ratio β as:

$$\beta_i = \left(\mu_i/\mu_1\right)^{1/m}, \quad i = 2, \ldots, K, \tag{21}$$

and, from (1), (2) and (3), $\mu_i \propto \left(\mathbf{d} \cdot \mathbf{n}_t^{(i)}\right)^m$, each β_i imposes

$$\mathbf{d} \cdot \left(\mathbf{n}_t^{(i)} - \beta_i \mathbf{n}_t^{(1)}\right) = 0, \quad i = 2, \ldots, K. \tag{22}$$

Denoting $\mathbf{a}_i = \mathbf{n}_t^{(i)} - \beta_i \mathbf{n}_t^{(1)}$, (22) reads $\mathbf{a}_i \cdot \mathbf{d} = 0$ and supplies $K-1$ homogeneous hyperplanes that intersect at the true $\mathbf{d}$.

2) Stochastic perturbation of the ratios: Let $P_{r,k}^{(i)}$ denote the k-th noisy observation gathered while the LED points along $\mathbf{n}_t^{(i)}$, $1 \le k \le N$, $1 \le i \le K$. According to (1), $P_{r,k}^{(i)} = \mu_i + \eta_i$ with $\eta_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d. The sample mean

$$\hat{\mu}_i = \frac{1}{N} \sum_k P_{r,k}^{(i)} \sim \mathcal{N}(\mu_i, \sigma^2/N)$$

is the MVU estimator. Injecting $\hat{\mu}_i$ into (21) yields

$$\hat{\beta}_i = g_i(\hat{\mu}_1, \hat{\mu}_i), \qquad g_i(x_1, x_2) = \left(\frac{x_2}{x_1}\right)^{\frac{1}{m}}.$$

A first-order Taylor expansion of g_i centred at $\boldsymbol{\mu}_i = (\mu_1, \mu_i)^\mathsf{T}$ gives

$$\hat{\beta}_i \approx \bar{\beta}_i + \frac{\bar{\beta}_i}{m}\left(\frac{\eta_i}{\mu_i} - \frac{\eta_1}{\mu_1}\right), \quad \bar{\beta}_i = (\mu_i/\mu_1)^{1/m}. \tag{23}$$

Hence every $\hat{\beta}_i$ shares the common perturbation η_1, producing correlation across orientations.

Define the ratio error $\tilde{\eta}_i = \hat{\beta}_i - \bar{\beta}_i$. Using (23) and the independence of η_i and η_1,

$$\mathrm{Var}[\tilde{\eta}_i] = \frac{\sigma^2}{Nm^2}\, \bar{\beta}_i^2 \left(\mu_i^{-2} + \mu_1^{-2}\right), \tag{24}$$

$$\mathrm{Cov}[\tilde{\eta}_i, \tilde{\eta}_j] = \frac{\sigma^2}{Nm^2}\, \bar{\beta}_i \bar{\beta}_j\, \mu_1^{-2}, \quad i \neq j. \tag{25}$$

The factor $(\mu_i^{-2} + \mu_1^{-2})$ in (24) demonstrates that the variance of $\tilde{\eta}_i$ depends explicitly on μ_i, even though each $\eta_k^{(i)}$ has the same variance σ^2. In other words, a single-LED system exhibits *heteroscedastic* ratio errors.

3) Residuals and their covariance.: Substituting $\hat{\beta}_i$ into $\mathbf{a}_i$, $\mathbf{a}_i = \bar{\mathbf{a}}_i - \mathbf{n}_t^{(1)} \tilde{\eta}_i$, and taking the inner product with $\mathbf{d}$ gives the residual

$$r_i = \mathbf{a}_i \cdot \mathbf{d} = \kappa\, \tilde{\eta}_i, \quad \kappa = \mathbf{n}_t^{(1)} \cdot \mathbf{d}.$$

Consequently, vector of residuals $\mathbf{r} = [r_2, \ldots, r_n]^\mathsf{T}$ is zero-mean Gaussian with covariance $\Sigma_r = \kappa^2 \Sigma_\beta$,

$$\mathbf{r} \;=\; A^\mathsf{T} \mathbf{d} \;+\; \varepsilon, \quad \varepsilon \sim \mathcal{N}\left(\mathbf{0}, \Sigma_r\right),$$

where Σ_β gathers (24)–(25). Higher-order remainders decay as $O(\sigma^4/N^2)$ and are negligible for $\mathrm{SNR} \ge 10$ dB and $N \ge 50$.

4) GLS formulation and closed-form solution: Collect the normals as $A = [\mathbf{a}_2 \;\ldots\; \mathbf{a}_n] \in \mathbb{R}^{3 \times (n-1)}$. With Σ_r known, the generalised least-squares problem is

$$\hat{\mathbf{d}} = \arg \min_{\|\mathbf{d}\|=1} \mathbf{d}^\mathsf{T} \left(A \Sigma_r^{-1} A^\mathsf{T}\right) \mathbf{d},$$

whose minimiser is the unit eigenvector associated with the smallest eigenvalue of $M_{\mathrm{GLS}} = A \Sigma_r^{-1} A^\mathsf{T}$. In practice, all multiplicative constants in Σ_r^{-1} can be omitted and absorbed into a normalisation of the weight matrix, since only the eigenvector direction matters. Formally,

$$\hat{\mathbf{n}}_{d,\mathrm{GLS}} = \frac{\mathbf{v}_{\min}\left(M_{\mathrm{GLS}}\right)}{||\mathbf{v}_{\min}\left(M_{\mathrm{GLS}}\right)||} \tag{26}$$

and the sign of $\hat{\mathbf{n}}_{d,\mathrm{GLS}}$ is chosen so that $\hat{\mathbf{n}}_{d,\mathrm{GLS}} \cdot \mathbf{n}_t^{(1)} > 0$, ensuring the vector points from the transmitter towards the receiver. This estimator is the minimum-variance, unbiased linear estimator (BLUE) for the linearised model.

B. WLS as a practical simplification

The inversion of Σ_r requires an $(n-1) \times (n-1)$ Cholesky factorisation. For real-time, low-power receivers a lighter alternative is attractive. Neglecting the off-diagonal terms in (25) gives the diagonal weight matrix

$$\tilde{\Sigma}_r = \mathrm{diag}\Big\{\bar{\beta}_i^2 \left(\mu_i^{-2} + \mu_1^{-2}\right)\Big\}_{i=2}^{n}.$$

Setting $\mathrm{W} = \tilde{\Sigma}_r^{-1}$ leads to the WLS cost

$$J_{\mathrm{WLS}}(\mathbf{d}) = \mathbf{d}^\mathsf{T} \left(AWA^\mathsf{T}\right) \mathbf{d}, \qquad \|\mathbf{d}\| = 1,$$

whose minimiser is

$$\hat{\mathbf{n}}_{d,\mathrm{WLS}} = \frac{\mathbf{v}_{\min}\left(AWA^\mathsf{T}\right)}{||\mathbf{v}_{\min}\left(AWA^\mathsf{T}\right)||}. \tag{27}$$

The neglected off-diagonal covariance terms scale as μ_1^{-2}, whereas the diagonal entries scale as $\mu_i^{-2} + \mu_1^{-2}$. When the reference orientation $(i = 1)$ satisfies $\mu_1 \gg \mu_i$ for most i, one obtains $\mu_1^{-2} \ll \mu_i^{-2}$, so that the off-diagonal terms are at least 20 dB below the diagonal terms. In such regimes, the WLS estimator captures over 95% of the statistical efficiency of GLS while avoiding matrix inversions. In contrast, in nearly symmetric geometries where $\mu_i \approx \mu_1$ for many i, off-diagonal correlations become important and the full GLS should be used for optimal performance.

VII. Results

Figure 6 presents the ~~cumulative distribution function~~ (CDF~~)~~ of the position estimation error for each algorithm introduced above, using optimal orientation sets of $K = 5$ and $K = 9$. The ~~generalized least-squares (~~GLS~~)~~ estimator achieves performance comparable to the ~~nonlinear (~~NL~~)~~ estimator and approaches the theoretical PEB, whereas the ~~weighted least-squares (~~WLS~~)~~ estimator incurs higher error but requires significantly lower computational complexity, thereby reducing latency.

Figure 7 compares the computation latency for $K = 5$ and $K = 9$. The NL estimator exhibits the greatest latency due

Algorithm 1 3-D Position Estimation from n Orientations

Require: Power samples $P_{r,k}^{(i)}$ ($k = 1, \dots, N$, $i = 1, \dots, K$); steering vectors $\mathbf{n}_t^{(i)}$; Lambertian order m; noise variance σ^2; MODE$\in \{\text{GLS}, \text{WLS}\}$; radiometric constant C.
Ensure: $\hat{\mathbf{R}}$ (3-D position of the receiver)

Direction finding:
  Compute $\hat{\mu}_i = \frac{1}{N}\sum_{k=1}^{N} P_{r,k}^{(i)}$ for $i = 1, \dots, K$
  Compute $\hat{\beta}_i = (\hat{\mu}_i/\hat{\mu}_1)^{1/m}$ for $i = 2, \dots, K$
  Form $\mathbf{a}_i = \mathbf{n}_t^{(i)} - \hat{\beta}_i\,\mathbf{n}_t^{(1)}$
  Set $A = [\mathbf{a}_2, \dots, \mathbf{a}_n]$
if MODE = GLS **then**
    Build Σ_r via (24)–(25), substituting $\beta_i \leftarrow \hat{\beta}_i$
    $M \leftarrow A\,\Sigma_r^{-1}\,A^{\mathsf{T}}$
else ▷ WLS (diagonal approximation)
    $\tilde{\Sigma}_r \leftarrow \operatorname{diag}\{\hat{\beta}_i^2(\hat{\mu}_i^{-2} + \hat{\mu}_1^{-2})\}$
    $M \leftarrow A\,\tilde{\Sigma}_r^{-1}\,A^{\mathsf{T}}$
end if
  Compute $\mathbf{v}_{\min}$ = eigenvector of M with smallest eigenvalue
  $\hat{\mathbf{n}}_d \leftarrow \mathbf{v}_{\min}/\|\mathbf{v}_{\min}\|$
if $\hat{\mathbf{n}}_d \cdot \mathbf{n}_t^{(1)} < 0$ **then**
    $\hat{\mathbf{n}}_d \leftarrow -\,\hat{\mathbf{n}}_d$ ▷ Ensure pointing to Rx
end if

Distance Estimation (beamsteering):
  Set $\mathbf{n}_t \leftarrow \hat{\mathbf{n}}_d$, $\mathbf{n}_r \leftarrow -\,\hat{\mathbf{n}}_d$
  Collect N power samples $P_r^{*(k)}$
  Compute $\hat{\mu}^* = \frac{1}{N}\sum_{k=1}^{N} P_{r,k}^*$
  $\hat{d} \leftarrow \sqrt{\frac{C}{\hat{\mu}^*}}$

Final 3-D position:
  $\hat{\mathbf{R}} \leftarrow \mathbf{T} + \hat{d}\,\hat{\mathbf{n}}_d$
return $\hat{\mathbf{R}}$

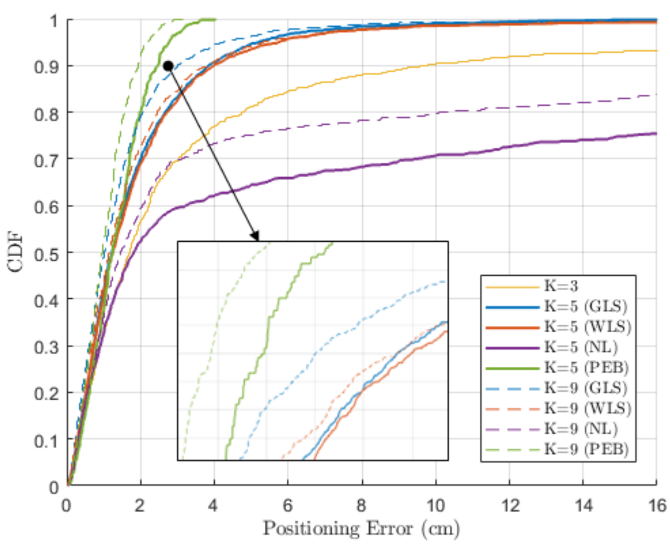


Fig. 6. Comparación de CDF de diferentes métodos y K.

to its iterative optimization procedure, while the closed-form GLS and WLS solutions execute more rapidly.

Figure 8 illustrates the 3D position estimates produced by WLS and GLS for $K = 5$ at three receiver heights $h = 0$, 0.6, and 1.2 m, plotted against the true reference positions. At $h = 1.2$ m, estimation errors increase near the testbed boundaries, particularly at the corners, consistent with the theoretical minimum PEB analysis (Fig. 3a) and attributable to reduced SNR at extreme incidence angles. In contrast, at $h = 0$ m the estimates exhibit greater spatial uniformity.

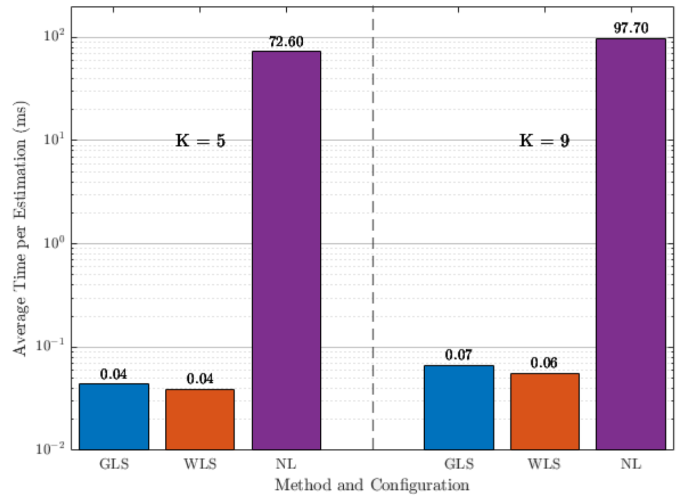


Fig. 7. Computation latency of the NL, GLS, and WLS estimators for $K = 5$ and $K = 9$.

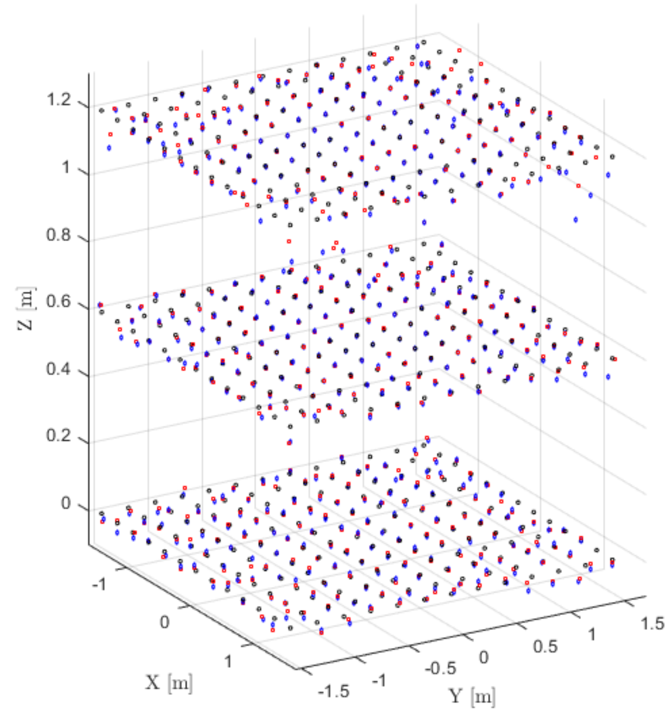


Fig. 8. 3D position estimates at three receiver heights for $K = 5$ orientations: GLS (blue), WLS (red), true positions (black).

VIII. Conclusion (draft)

This paper presented a novel beam-steered single-LED VLP system for accurate 3D positioning. The key contributions include: (1) a comprehensive mathematical framework incorporating both linear and nonlinear estimators, (2) derivation of the theoretical PEB for performance analysis, (3) genetic algorithm-based optimization of LED orientation sets, and (4) extensive performance evaluation.

The proposed system achieves centimeter-level accuracy while requiring minimal infrastructure compared to traditional multi-LED systems. The optimized 5-orientation configuration provides an excellent trade-off between positioning accuracy and system complexity. Future work will explore real-time implementation and experimental validation of the proposed algorithms.

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